TEXORA · LECTURE SHEET

Electric Network Fundamentals

Ohm’s law and Kirchhoff’s laws; branch-current, mesh, superposition and Thevenin methods of network analysis; sinusoidal single-phase circuits with R, L and C in series and parallel; real and reactive power; and power factor with the methods of improving it.

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 1 · LEVEL 2 TERM 2

1. Ohm’s law and basic network quantities

An **electric network** is any interconnection of sources and elements (resistors, inductors, capacitors) through which current can flow. The part of a network between two junctions is a **branch**; a junction where three or more branches meet is a **node**; a closed conducting path is a **loop**, and a loop containing no other loop inside it is a **mesh**.

Ohm’s law states that, at constant temperature, the current through a conductor is directly proportional to the voltage across it:

$$V = IR \quad \cdot \quad P = VI = I^2R = V^2/R$$

where V is voltage (volt), I current (ampere), R resistance (ohm, Ω) and P the power converted to heat (watt). Resistors in **series** add: $R = R_1 + R_2 + \dots$ (same current through each). Resistors in **parallel** add reciprocally: $1/R = 1/R_1 + 1/R_2 + \dots$ (same voltage across each); for two resistors, $R = R_1R_2/(R_1 + R_2)$.

2. Kirchhoff’s laws

Kirchhoff’s current law (KCL) — the algebraic sum of currents at any node is zero: currents entering equal currents leaving. It is simply conservation of charge; charge cannot accumulate at a junction.

$$\Sigma I \text{ (entering)} = \Sigma I \text{ (leaving)}$$

Kirchhoff’s voltage law (KVL) — the algebraic sum of voltages around any closed loop is zero: the sum of EMF rises equals the sum of voltage drops. It is conservation of energy: a charge carried round a loop returns to its starting potential. Sign convention used here: traversing a resistor in the direction of its current gives a *drop* IR; traversing a source from − to + gives a *rise* E.

3. The branch-current method

Steps: (1) label every branch current with an assumed direction; (2) write KCL at a node to relate the branch currents; (3) write KVL for as many independent loops as needed; (4) solve the simultaneous equations. A negative answer means the true direction is opposite to the one assumed.

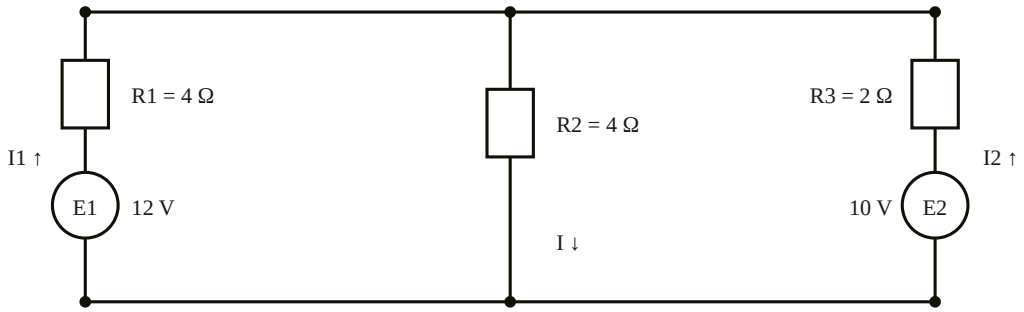


Fig. 1 — Two sources feeding a common load R2; branch currents I1, I2 and I as assumed.

WORKED EXAMPLE — BRANCH CURRENTS

For Fig. 1: KCL at the top node: I = I1 + I2. Left loop: 12 = 4I1 + 4I. Right loop: 10 = 2I2 + 4I. Trying I1 = 1 A, I2 = 1 A gives I = 2 A; check: left loop 4(1) + 4(2) = 12 ✓; right loop 2(1) + 4(2) = 10 ✓.

So **I1 = 1 A**, **I2 = 1 A**, and the load carries **I = 2 A** (24 W dissipated in R2).

4. The mesh (loop-current) method

Instead of branch currents, assign one circulating current to each mesh (conventionally clockwise). A branch shared by two meshes carries the difference of the two mesh currents. For each mesh, write KVL in the form: (sum of resistances in the mesh) × (its mesh current) − Σ (shared resistance × neighbouring mesh current) = (sum of EMF rises in the clockwise direction).

For Fig. 1, with mesh currents J1 (left mesh) and J2 (right mesh), both clockwise:

$$8J1 - 4J2 = 12 \;\cdot\; -4J1 + 6J2 = -10$$

Solving: J1 = **1 A**, J2 = **−1 A**. The negative sign means the right mesh current actually circulates anticlockwise — source E2 is supplying current upward through its branch, exactly as the branch method found: load current = J1 − J2 = 1 − (−1) = 2 A downward. The mesh method needs fewer equations than the branch method on large networks, which is why it is preferred for systematic analysis.

5. The superposition theorem

Statement: in a linear network containing several independent sources, the current in any branch is the algebraic sum of the currents produced by each source acting *alone*, with all other voltage sources short-circuited (replaced by their internal resistance) and all current sources open-circuited. It applies only to linear networks and to currents and voltages — never to power, which depends on the square of current.

WORKED EXAMPLE — SUPERPOSITION

Circuit: E1 = 12 V in series with R1 = 3 Ω feeds a node; from the node, R2 = 6 Ω goes to the common rail, and a second branch has E2 = 6 V in series with R3 = 3 Ω (both sources positive at the top). Find the current through R2.

E1 alone (E2 shorted): resistance seen = R1 + (R2 ‖ R3) = 3 + 2 = 5 Ω; total current 12/5 = 2.4 A; through R2 by current division: 2.4 × 3/(6 + 3) = **0.8 A** downward.

E2 alone (E1 shorted): resistance seen = R3 + (R2 ‖ R1) = 3 + 2 = 5 Ω; total 6/5 = 1.2 A; through R2: 1.2 × 3/(3 + 6) = **0.4 A** downward.

Total through R2 = 0.8 + 0.4 = **1.2 A** downward. (Check by nodal analysis: node voltage = 7.2 V, and 7.2/6 = 1.2 A ✓.)

6. Thevenin’s theorem

Statement: any linear network containing sources and resistors, viewed from two terminals, behaves exactly like a single voltage source V_{th} in series with a single resistance R_{th}. V_{th} is the open-circuit voltage at the terminals; R_{th} is the resistance measured at the terminals with all independent sources turned off (voltage sources shorted, current sources opened). The load current is then $I_L = V_{th}/(R_{th} + R_L)$. (The dual form, a current source $I_N = V_{th}/R_{th}$ in parallel with R_{th}, is Norton’s equivalent.)

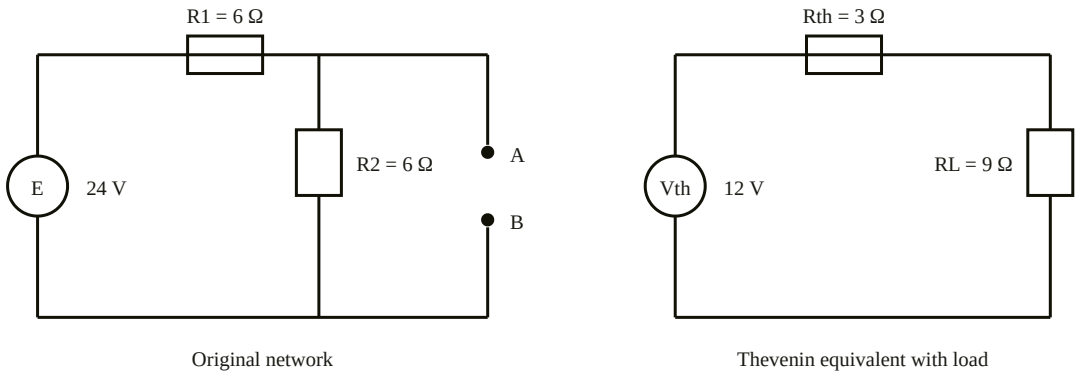


Fig. 2 — A network and its Thevenin equivalent at terminals A–B.

WORKED EXAMPLE — THEVENIN

For Fig. 2: V_{th} = voltage across R2 with A–B open = 24 × 6/(6 + 6) = **12 V**. R_{th} = R1 ‖ R2 = 6 ‖ 6 = **3 Ω**. With RL = 9 Ω connected: I_L = 12/(3 + 9) = **1 A**; power in the load = I^2RL = 9 W.

> Maximum power transfer occurs when RL = R_{th}.

7. Sinusoidal single-phase A.C.: the three elements

A sinusoidal voltage $v = V_m \sin \omega t$ ($\omega = 2\pi f$) drives a current whose phase relative to the voltage depends on the element. Using phasors (RMS values V and I), the opposition each element offers is:

ELEMENT	OPPOSITION	PHASE OF CURRENT VS VOLTAGE
Resistor R	R (Ω)	In phase
Inductor L	Reactance XL = ωL = 2πfL	Lags by 90°
Capacitor C	Reactance XC = 1/(ωC) = 1/(2πfC)	Leads by 90°

Only the resistor consumes average power. The inductor and capacitor alternately store energy (in a magnetic field and an electric field respectively) and return it to the supply each cycle.

8. The series R–L–C circuit and resonance

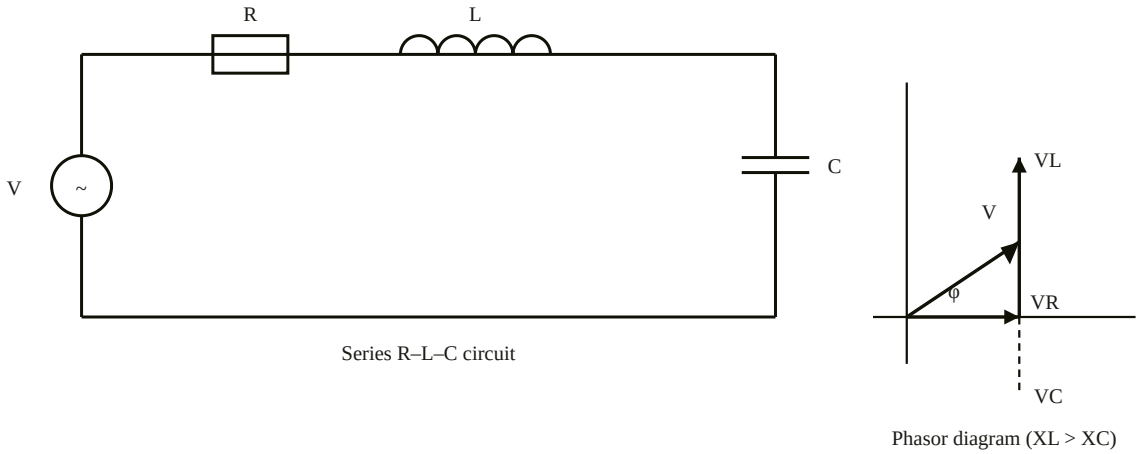


Fig. 3 — Series R–L–C circuit and its phasor diagram; V is the phasor sum of VR, VL and VC.

The same current flows through all three elements. The **impedance** is

$$Z = \sqrt{(R^2 + (XL - XC)^2)}, \quad I = V/Z, \quad \tan \varphi = (XL - XC)/R$$

where φ is the angle by which the current lags the voltage (positive when XL > XC, inductive) or leads it (capacitive). **Series resonance** occurs when XL = XC, at $f_0 = 1/(2\pi\sqrt{(LC)})$: the impedance is then a minimum (= R), the current a maximum, and the power factor unity. For L = 0.1 H and C = 10 μF, $f_0 = 1/(2\pi \times 10^{-3}) =$ **159 Hz**.

> **WORKED EXAMPLE — SERIES R–L–C**

R = 40 Ω, XL = 80 Ω, XC = 50 Ω on a 230 V supply. $Z = \sqrt{(40^2 + (80 - 50)^2)} = \sqrt{(1600 + 900)} =$ **50 Ω**. $I = 230/50 =$ **4.6 A**. Power factor = R/Z = 0.8, lagging (the circuit is net inductive). Real power $P = I^2R = 4.6^2 \times 40 =$ **846.4 W**; reactive power $Q = I^2(XL - XC) = 4.6^2 \times 30 = 634.8$ var; apparent power $S = VI = 1058$ VA. Check: $\sqrt{(846.4^2 + 634.8^2)} = 1058$ ✓.

9. The parallel R–L–C circuit

In parallel, the voltage is common and the branch currents add as phasors. It is easier to work with **admittance** $Y = 1/Z = \sqrt{(G^2 + (BC - BL)^2)}$, where $G = 1/R$ is the conductance and $BC = \omega C$, $BL = 1/(\omega L)$ are the susceptances. **Parallel resonance** occurs when BC = BL, again near $f_0 = 1/(2\pi\sqrt{(LC)})$: the susceptances cancel, the admittance is a minimum (= G), so the impedance is a *maximum* and the line current a minimum at unity power factor — the dual of the series case. A practical coil (L with resistance) in parallel with C behaves the same way to a first approximation and is the basis of tuned "tank" circuits.

FEATURE	SERIES RESONANCE	PARALLEL RESONANCE
Condition	XL = XC	BC = BL
Impedance	Minimum (= R)	Maximum
Line current	Maximum	Minimum
Power factor	Unity	Unity
Common name	Acceptor circuit	Rejector circuit

10. Power in A.C. circuits and power factor

With v and i sinusoidal and φ the phase angle of the current relative to the voltage:

$$\text{Real power } P = VI\cos\varphi \text{ (W)} \cdot \text{ Reactive power } Q = VI\sin\varphi \text{ (var)} \cdot \text{ Apparent power } S = VI \text{ (VA)}, \; S^2 = P^2 + Q^2$$

Only P does useful work; Q shuttles back and forth, loading generators, transformers and cables without delivering energy. The **power factor**, $\cos\varphi = P/S =$ (kW)/(kVA), measures how effectively the drawn current is being used. Most industrial loads (induction motors, fluorescent fittings) are inductive and lagging.

Disadvantages of low power factor: for the same real power the line current is larger, so cables and windings must be bigger, I^2R losses are higher, voltage regulation worsens, and the capacity of generators and transformers is under-used; supply authorities therefore penalise consumers whose

power factor is poor. **Improvement methods:** (1) connect **static capacitors** (or a capacitor bank) in parallel with the load — the commonest and cheapest method, since the capacitor supplies the lagging reactive power locally; (2) run a **synchronous condenser** (an over-excited synchronous motor) at large installations; (3) use phase advancers for induction motors; and avoid running motors lightly loaded, which itself worsens the power factor.

WORKED EXAMPLE — POWER-FACTOR IMPROVEMENT

A 230 V, 50 Hz single-phase motor takes 10 A at power factor 0.6 lagging. Find the capacitor needed to raise the power factor to 0.9 lagging, and the new line current.

Real power $P = VI\cos\varphi_1 = 230 \times 10 \times 0.6 = 1380$ W (unchanged by correction).

$\varphi_1 = \cos^{-1} 0.6 = 53.13°$, $\tan\varphi_1 = 1.333$; $\varphi_2 = \cos^{-1} 0.9 = 25.84°$, $\tan\varphi_2 = 0.484$.

Capacitor reactive power $Q_C = P(\tan\varphi_1 - \tan\varphi_2) = 1380 \times (1.333 - 0.484) =$ **1172 var ≈ 1.17 kvar**.

$C = Q_C/(2\pi fV^2) = 1172/(2\pi \times 50 \times 230^2) = 7.05 \times 10^{-5}$ F ≈ **70 μF**.

New line current $= P/(V\cos\varphi_2) = 1380/(230 \times 0.9) =$ **6.67 A** — the same 1380 W is now delivered with a third less current.

11. Key terms

- **Node / branch / mesh:** a junction of three or more branches / a single path between junctions / a closed loop with no loop inside it.
- **Linear network:** one whose element values do not change with voltage or current, so superposition applies.
- **Thevenin voltage V_{th}:** the open-circuit voltage at a pair of terminals; R_{th} is the resistance looking back into them with sources turned off.
- **Reactance:** the frequency-dependent opposition of a pure inductor ($X_L = \omega L$) or capacitor ($X_C = 1/(\omega C)$), in ohms.
- **Impedance Z:** the total phasor opposition of an A.C. circuit; admittance $Y = 1/Z$.
- **Resonance:** the condition $X_L = X_C$ at which a circuit's reactive effects cancel.
- **Power factor:** $\cos\varphi = P/S$; lagging when current lags voltage (inductive load), leading when it leads (capacitive load).

12. Key points

- KCL is charge conservation at a node; KVL is energy conservation round a loop. Every network method is built on them.

- Mesh analysis assigns one current per mesh and writes KVL in a standard pattern — the fastest hand method for multi-loop networks.
- Superposition: one source at a time, others turned off; add currents and voltages algebraically, never powers.
- Thevenin reduces any linear network at two terminals to V_{th} in series with R_{th}; load current = $V_{th}/(R_{th} + RL)$.
- Series RLC: $Z = \sqrt{(R^2 + (XL - XC)^2)}$; resonance at $f_0 = 1/(2\pi\sqrt{(LC)})$ makes Z minimum and current maximum. In parallel resonance the impedance is maximum and the line current minimum.
- Only resistance consumes average power: $P = VI \cos \varphi$. Low power factor means excess current for the same work; parallel capacitors are the standard cure.

13. Summary

D.C. networks yield to Kirchhoff's two laws, applied directly (branch currents), systematically (mesh currents), or one source at a time (superposition); Thevenin's theorem compresses any such network into a single source and resistance at the terminals of interest. In sinusoidal circuits the same ideas work with phasors: resistance, inductive reactance and capacitive reactance combine into an impedance, resonance occurs when the reactances cancel, and the power drawn splits into a working part $P = VI \cos \varphi$ and a circulating part $Q = VI \sin \varphi$. Keeping the power factor high — normally by adding parallel capacitance — reduces current, losses and equipment size for the same useful power.

14. Practice

Answer in your own words or by calculation; brief answers follow. No multiple-choice questions.

1. A 24 V source feeds R1 = 6 Ω in series with a parallel pair R2 = 12 Ω and R3 = 6 Ω. Find the total current from the source and the current through R3.
2. For a network, V_{th} = 8 V and R_{th} = 1.6 Ω. What load current flows when RL = 8 Ω is connected? (Hint: find V_{th} and R_{th} for E = 10 V, a series 2 Ω resistor and an 8 Ω resistor across the terminals, and verify your answer matches.)
3. A series R–L circuit has R = 30 Ω and XL = 40 Ω on a 100 V supply. Find the current, the power factor and the power consumed.
4. Find the resonant frequency of a series circuit with L = 0.05 H and C = 20 μF, and state the impedance and power factor at resonance if R = 10 Ω.
5. Explain, with one reason each, why (a) superposition cannot be used to add powers, and (b) a factory is charged a penalty for running at a power factor of 0.6.

Answer key

1. Parallel pair: (12 × 6)/(12 + 6) = 4 Ω. Total R = 6 + 4 = 10 Ω; total current = 24/10 = 2.4 A. Voltage across the pair = 2.4 × 4 = 9.6 V; current through R3 = 9.6/6 = 1.6 A.
2. $I_L = 8/(1.6 + 8) = 0.83$ A. (From the data: $V_{th} = 10 \times 8/(2 + 8) = 8$ V; $R_{th} = 2 \parallel 8 = 1.6$ Ω ✓.)
3. $Z = \sqrt{(30^2 + 40^2)} = 50$ Ω; I = 100/50 = 2 A; power factor = R/Z = 0.6 lagging; $P = I^2R = 4 \times 30 = 120$ W.
4. $f_0 = 1/(2\pi\sqrt{(LC)}) = 1/(2\pi\sqrt{(0.05 \times 20 \times 10^{-6})}) = 1/(2\pi \times 10^{-3}) = 159$ Hz. At resonance Z = R = 10 Ω (minimum) and the power factor is unity.
5. (a) Power depends on the square of current ($P = I^2R$), and $(I1 + I2)^2 \neq I1^2 + I2^2$, so the linear addition used for currents fails for power. (b) At 0.6 power factor the factory draws about 1/0.6 ≈ 1.67 times the current needed for its real power, increasing the supplier's cable losses and tying up generator and transformer capacity; the penalty pays for that wasted capacity.

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